# ETHANOL AS A TRANSPORTATION FUEL IN INDIA — INSTITUTIONAL RESEARCH FOUNDATION

*Analytical backbone for a sell-side thematic equity report. No individual tickers (out of scope). Facts/estimates/opinions are labeled throughout. "Reported" = official/primary data; "Estimated" = modeled/derived via first principles; "Opinion" = author judgment for an institutional investor. Data flags note where figures are self-stated by government and not independently audited.*

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## SECTION 1 — THE BIG STRATEGIC QUESTION

**Core question:** Can ethanol be a durable long-term substitute for petrol, or is it a transitional bridge before EVs dominate?

**Author's most-likely conclusion (Opinion):** Ethanol blending in India is a **durable but capped theme** — a structural winner up to ~E20–E27 over the next 10–15 years, but **NOT a path to majority petrol substitution and not a petrol "replacement."** It is best understood as: (a) a powerful petrol-demand *moderator* and farm-income/energy-security instrument; (b) a *transitional* technology in two-wheelers and passenger cars where EVs will progressively take share post-2030; and (c) structurally constrained by water, food-vs-fuel and land limits beyond ~E20–E30. The most aggressive ambitions (E85/E100 mass fleets) are unlikely to succeed at scale. Crucially, ethanol addresses only the **petrol pool (~40 MMT)**, not India's larger **diesel pool (~91 MMT)** — so even total success caps its impact on overall oil imports.

**Bull case (rigorous):**

- India imports ~88% of crude; ethanol is domestically produced → energy security + trade-balance relief.
- Reuses the existing ICE fleet + fuelling infrastructure → far lower capital intensity than full electrification.
- Politically durable farm-income transfer mechanism (no government has reversed blending targets in over a decade).
- Octane benefit (blending RON ~120–135) enables higher compression ratios → efficiency offset in tuned engines.

- Sugarcane ethanol delivers genuine lifecycle CO2 reductions.

**Bear case (rigorous):**

- The energy-density penalty is physics, not engineering: ethanol carries ~65% of petrol's volumetric energy → unavoidable mileage loss.
- Water intensity is severe (NITI Aayog: ~2,860 L water per L of sugarcane ethanol; ~10,790 L for rice-based).
- Food-vs-fuel: the pivot to maize triggered import dependence and feed-price inflation.
- Subsidy/administered-price dependence; OMCs are forced to buy ethanol at prices often above petrol's pre-tax base.
- EVs are improving on the cost curve; ethanol's "transitional" window may close faster than capacity payback.
- The blend wall: ~20% is the practical ceiling for the legacy fleet; higher blends need flex-fuel vehicles (FFVs) that face 18–40% GST vs 5% for EVs.

**Do NOT assume government policy is correct (Opinion):** The program has migrated from its original purpose — clearing the **sugar surplus** and helping mills pay cane farmers — to a **grain-based (maize/rice) industry**. This structural change undermines the original "waste-to-value" and water-efficiency rationale and introduces a direct food-security tension.

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## SECTION 2 — GLOBAL ETHANOL EXPERIENCE (most important section)

### BRAZIL — the global benchmark

- **History (Reported):** Proálcool launched 1975 post-oil-shock. A mandatory blend was fixed by a 1993 law at E22; 2003 rules set a min E20/max E25 band; raised to **E25 from July 2007**; the lower limit was cut to E18 in April 2011 during a supply shortage; **raised to E27 in March 2015** specifically to clear an ethanol overstock of over 1 billion litres. First E100 car 1979; by 1986, E100 vehicles were 92% of the light-vehicle market. Production rose from 0.6 bn L (1975) to 12.7 bn L (1992).
- **Flex-fuel revolution (Reported):** FFVs launched 2003; by 2009 ~90% of new car sales. Today ~73% of the fleet (~27 million vehicles) can use ethanol. Flex motorcycles passed 4 million units by 2015.
- **Current blend (Reported):** Confirmed by S&P Global Platts, the new **E30 mandate**

**took effect August 1, 2025**, requiring all Gasoline C to contain 30% anhydrous ethanol (up from 27.5%), set by CNPE Resolution No. 9/2025 under the "Fuel of the Future" Law 14.993/2024 — which permits raising the blend up to **35%** on proof of technical viability. Hydrous E100 is sold alongside for flex vehicles. Consumers choose at the pump on the ~70% price-parity rule (ethanol attractive when its price is <70% of gasoline, reflecting energy content). USDA

- **Economics (Reported):** Sugarcane energy balance 8–10x (output/input); bagasse cogeneration makes mills energy self-sufficient. Wikipedia Land use: sugarcane for all purposes ~8 million ha (~1% of land / ~2.4% of arable). ScienceDirect Average yield ~75 t/ha, ~7,000 L ethanol/ha.
- **Can it be replicated in India? (Opinion — critical): Only partially.** Brazil's advantages are structurally absent in India: (1) Brazil has abundant arable land and water; India is water-stressed, with cane+paddy already using ~70% of irrigation water. (2) Brazilian consumers get *price choice and discounts at the pump*; Indians get E20 only, at the same or higher price. (3) Brazil built FFVs over 20+ years; India's FFV fleet is nascent with punitive GST. (4) Brazil's cane yields and 8–10x energy balance are world-leading; India's are lower. **Verdict: India can reach E20–E30 but cannot replicate Brazil's E100/flex consumer model at scale.**

### UNITED STATES — corn ethanol & the blend wall

- **Reported:** RFS (2005, expanded 2007 EISA) mandated rising volumes to 36 bn gallons by 2022, capped at 15 bn gal corn-starch ethanol. E10 is ubiquitous (the de facto national standard). E15 is approved for MY2001+ vehicles but uptake is slow. E85 serves ~21 million FFVs but is limited (~4,300 of ~196,000 stations).
- **Subsidy history (Reported):** The VEETC blender's credit (~$6 bn/yr) expired end-2011; corn ethanol now runs on the RFS mandate + RIN credits + corn subsidies (>30% of the US corn crop → ethanol), with a new 45Z clean-fuel credit layered on.
- **Key insight (Opinion):** Even the world's largest ethanol producer is **stuck at the E10 blend wall.** AFPM analysis argues E10 would dominate even without the RFS, and pushing E15/E85 costs ~$770 per incremental ethanol gallon. The US proves that, without flex fleets and pump-price choice, ethanol plateaus at ~10%.

### EUROPE — deliberately limited

- **Reported:** E5/E10 standard; 19 EU countries + Norway + UK have E10. E10 is still below its 10% blend wall — Germany ~26%, France ~58% E10 market share. RED III targets a 28.5% renewable-transport share / 13% GHG-intensity cut by 2030, with a 7% cap on first-generation (food-crop) biofuels. ScienceDirect E20 is only now being normalized in

CEN (~2026).

- **Why limited (Opinion):** Europe chose **electrification** as its primary decarbonization path and treats crop-based biofuels with sustainability suspicion (indirect land-use-change concerns). Ethanol is a marginal complement, not a strategy.

### CHINA, THAILAND, JAPAN & others

- **China (Reported):** Announced a nationwide E10 mandate (2017) targeting 2020. International Energy Agency **Failed/rolled back** — the blend rate never exceeded ~4%, with reports of lowering ambition to E5. Ethanol Producer Magazine Grain-stock dynamics and food-security priorities derailed it.

- **Thailand (Reported):** Successful demand stimulation via subsidies (E10/E20/E85 gasohol, ~13.8% blend in 2018), but a **2024 reversal:** phasing out E85 and gasohol 91/95, consolidating on E20 by 2027, with surplus ethanol (50%+ over road demand, 28 plants / 6.92 MLPD capacity vs ~3.43 MLPD road use) redirected to Sustainable Aviation Fuel. Advanced BioFuels USA Subsidies became fiscally unsustainable under the State Oil Fund Act. USDA

- **Japan (Reported):** Prefers **ETBE** (ethanol→ETBE) over direct blending — the refiners' preferred, conservative route with low direct-ethanol blend. Acfa

- **Lesson for India (Opinion):** China's failure and Thailand's retreat from high blends are cautionary: high-blend ambitions collide with food security (China) and fiscal sustainability (Thailand).

### The critical question: WHY haven't advanced markets gone to very high blends?

**(Opinion, evidence-backed):** Because (1) the **blend wall** — legacy fleets and infrastructure cap practical blending at ~10–20% without dedicated FFVs; (2) the **energy-density penalty** is unavoidable; (3) **subsidy/fiscal cost** scales unfavorably (US $770/incremental gallon; Thailand's subsidy burden); (4) **food-vs-fuel politics** (China); and (5) advanced markets chose **electrification** as the deeper decarbonization bet. Only Brazil — with unique land/water/yield endowments and a 40-year flex fleet — sustains high blends. **This is the single most important fact challenging India's E85/E100 ambitions.**

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## SECTION 3 — INDIA'S ETHANOL JOURNEY

**Reported timeline:**

- EBP launched 2001 (pilot, 5% in 9 states); blending ~1.5% by ESY 2013-14.
- National Policy on Biofuels 2018: E20 by 2030; multiple feedstocks allowed.
- June 2021: E20 target advanced to 2025; NITI Aayog "Roadmap for Ethanol Blending in India 2020-25" released; E100 pilot launched in Pune.
- 2022: 10% blending achieved (ESY 2021-22), ahead of schedule; NPB amended to formally advance E20 to 2025-26. GST on ethanol cut 18%→5%; Drishti IAS 6% interest subvention for distilleries.
- 2023: Global Biofuels Alliance launched at G20.
- **ESY 2024-25 blending: 19.2% cumulative; 20% reached in the month of October 2025** (PPAC). OMCs received 1,003.1 cr L and blended 1,022.4 cr L — crossing 1,000 cr L for the first time.
- **April 2026: E20 effectively the standard petrol grade nationwide, with min RON 95** (BIS). E0 largely phased out. The Supreme Court (Sep 1, 2025) dismissed a PIL challenging mandatory E20 sale.
- **BIS IS 19850:2026** notified specifications for E22/E25/E27/E30; E27 guidelines expected ~Aug 2026.

**Rationale (Reported):** Reduce crude imports/forex; energy security; farmer income; emissions; trade balance.

**Government benefit claims (Reported — self-stated, DATA FLAG: not independently audited):** Per MoPNG, as of July 2025 (cumulative from ESY 2014-15) the program "saved over Rs. 1.44 lakh crore in foreign exchange, substituted 245 lakh metric tonnes of crude oil, and reduced CO2 emissions by approximately 736 lakh metric tonnes" (DD News, Aug 2025). Petroleum Minister Hardeep Singh Puri later cited (Business Standard, June 4, 2026) "₹1.84 lakh crore in foreign exchange... 909 lakh metric tonnes reduction in $CO_2$ emissions... Rs 1.58 lakh crore earnings to farmers." These figures escalate across statements and should be treated as policy claims, not verified accounting.

**Is the trajectory sustainable? (Opinion):** E20 is achieved and supply-sustainable. Beyond E20, sustainability is **questionable** — installed capacity already exceeds demand (oversupply risk), and pushing to E25–E30 requires either more cane-juice diversion (sugar-security tension) or more maize/rice (food-feed tension). The constraint has flipped from supply-limited to demand-limited.

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## SECTION 4 — ENGINEERING & AUTOMOTIVE PERSPECTIVE

**Fuel properties (Reported — IEA-AMF, US DOE/AFDC):**

| Property | Gasoline | Ethanol |
|---|---|---|
| Lower heating value (MJ/kg) | ~42.5 | ~26.8–28.9 |
| Volumetric energy density (MJ/L) | ~32 | ~21 (≈65% of gasoline) |
| RON | 92–98 | ~129 (blending RON 120–135) |
| MON | 82–88 | ~102 (blending 100–106) |
| Oxygen content | 0% | 35% |
| Stoichiometric A/F ratio | 14.6 | 9.0 |
| Density (kg/L) | ~0.74 | ~0.79 |

- **Energy density / mileage (Reported + Estimated):** Ethanol's volumetric energy is ~33% lower. ResearchGate Theoretical fuel-consumption rise for E10 vs non-oxygenated gasoline is ~3.5%. IEA AMF For E20: ARAI/MoPNG cite a **1–2% mileage drop for E10-designed/E20-calibrated four-wheelers and 3–6% for others.** Energy-content math implies ~4% less energy per litre at E20; the gap between ~4% theoretical and "1–2%" official is partly recovered via octane/calibration but is contested. **Real-world (Reported):** A LocalCircles survey released February 2026 (50,000+ responses across 301 districts) found "1 in 2 petrol vehicle owners with a vehicle purchased in 2022 or earlier say their fuel efficiency/mileage has reduced," and 29% confirmed experiencing unusual wear/tear or repair needs. (A separate October 2025 LocalCircles wave found these figures rising to ~80% reporting mileage decline and ~52% reporting wear — DATA FLAG: survey/self-reported, not instrumented.)
- **Octane advantage (Reported):** High RON enables higher compression ratios; in optimized engines this can partially or fully offset the calorific penalty (km/MJ can exceed gasoline). IEA AMF BUT this requires *dedicated E20-tuned engines*, not the legacy fleet.
- **Emissions (Reported/Estimated):** Oxygenated combustion → lower CO and HC. But higher **aldehyde** (acetaldehyde) emissions, potential NOx shifts, and evaporative/RVP issues at mid blends. Lifecycle $CO_2$ depends entirely on feedstock — sugarcane good, maize ~2x sugarcane GHG, rice worst (water/methane).
- **Material compatibility (Reported):** Ethanol is hygroscopic (water absorption → phase separation, corrosion) and degrades elastomers/rubber/certain plastics and corrodes aluminium/fuel-system components in non-compatible vehicles. Per ARAI/SIAM, vehicles

were designed for E5 with E10-compatible rubber/plastic. E20 material-compliant + E10-tuned vehicles rolled out from April 2023; E20-tuned engines from April 2025 (BS6 Phase 2).

- **Cold start (Reported):** Ethanol's higher heat of vaporization → harder cold starts; Brazil solved this via a secondary gasoline tank, then Bosch "Flex Start" fuel-heating (cold start to –5°C).
- **E20 modification cost / OEM margin (Opinion + Reported):** Incremental cost is modest for new vehicles (material upgrades + ECU calibration), largely absorbed and already implemented from April 2023. The real burden falls on the **legacy fleet** (pre-2020/2023 vehicles), which cannot be retrofitted economically — a cost transferred to consumers, not OEMs.
- **E85/E100 in India (Opinion):** Practical only with dedicated FFVs; faces GST disadvantage and infrastructure gaps. Unlikely to scale beyond niche within the decade.

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## SECTION 5 — CONSUMER PERSPECTIVE

- **Mileage / running cost (Estimated + Reported):** ~1–6% mileage loss (official) vs 5–12% (user-reported). Critically, **E20 is sold at the same or higher price as regular petrol** despite lower energy. NITI Aayog's 2021 roadmap explicitly recommended that the "retail price of such fuels should be lower than normal petrol to compensate for the reduction in calorific value" Substack — a recommendation **ignored** in practice. Net effect: consumers pay premium prices for lower-energy fuel.
- **No fuel choice (Reported):** Unlike Brazil, Indian consumers cannot choose blend at the pump; E0/low-ethanol is increasingly unavailable or premium-priced.
- **Maintenance (Reported):** Legacy-vehicle owners report corrosion, seal/gasket degradation, and injector-wear concerns.
- **Ethanol vs hybrid vs EV preference (Opinion):** For new buyers the choice is an E20 ICE (no upfront premium), a hybrid (efficiency, no range anxiety), or an EV (low running cost, 5% GST, but charging/upfront cost). Ethanol-ICE wins on upfront cost/convenience today; EVs win long-run TCO in 2W/3W; hybrids occupy the pragmatic PV middle.

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## SECTION 6 — DETAILED QUANTITATIVE ANALYSIS (bottom-up, with assumptions)

**Current fuel ecosystem (Reported — PPAC, FY2024-25):**

- Total petroleum product consumption: **239.2 MMT**.
- Crude import dependence: **88.2%**; crude imports **234.3 MT**; **gross crude import bill ~USD 137 billion** (PPAC). *DATA FLAG: ICRA's forward-looking net oil-import bill estimate was ~$101–104 bn; the $137bn is gross crude per PPAC.*
- Diesel (HSD): **91.4 MMT** (~38–40% of slate); Petrol (MS): **40.0 MMT** (~16–17%). **Diesel:petrol consumption ratio ~2.3:1.**
- Petrol by segment (Reported — PPAC/Nielsen study): **Two-wheelers 61.4%, cars 34.3% (passenger cars ~30% + UVs ~10% on retail-only cut), three-wheelers 2.3%.** 99.6% of petrol goes to transport.

**Ethanol replacement math (First-principles calculation; assumptions stated):**

- Assume petrol ~40 MMT/yr, density ~0.74 kg/L → 40,000,000 t ÷ 0.74 = **~54 billion litres of petrol/yr** (Estimated).
- Ethanol required by blend (volume basis):

| Blend | Ethanol needed (bn L) | Note |
|---|---|---|
| E10 | ~5.4 | |
| E20 | ~10.8 | ≈ NITI's 1,016 cr L (10.16 bn L) blending requirement |
| E50 | ~27 | |
| E85 | ~46 | requires FFV fleet |
| E100 | ~54 | full volumetric replacement |

- **Cross-check vs NITI (Reported):** The NITI Aayog Roadmap (June 2021) projected India's ethanol requirement for petrol blending rising "from 173 crore litres in 2019-20 to 1,016 crore litres in 2025-26," with production capacity needing to grow "from 684 crore litres in 2019-20 to 1,500 crore litres in 2025-26" — consistent with the ~10.8 bn L E20 figure above.
- **Energy-adjusted reality (critical, Estimated):** Because ethanol holds ~65% of petrol's volumetric energy, replacing 100% of petrol *energy* would need ~54 ÷ 0.65 ≈ **83 billion litres** — far more than the volumetric figure implies. **E20 (20% by volume) displaces only ~13% of petrol energy.**
- **What % is realistically substitutable? (Opinion):** Volumetrically, E20 is achieved;

E25–E30 is feasible with strain. **Complete replacement (E100) is not realistic** — it would need ~83 bn L vs current capacity ~18 bn L and consume an impossible share of land/water/grain. **Realistic durable ceiling: ~20–30% volumetric (~13–20% energy) for the blended fleet.**

**Chart idea 6A:** Stacked bar — ethanol volume required (bn L) at E10/E20/E50/E85/E100, with a horizontal reference line for current installed capacity (~18 bn L / ~1,800 cr L). *Underlying data = table above.* **Chart idea 6B:** "Volumetric blend % vs actual petrol-energy displaced %" dual bars (E10→3.5%×… ; E20→~13%; E85→~55%). **Chart idea 6C:** Pie — petrol consumption by segment (2W 61.4% / cars 34.3% / 3W 2.3% / other).

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## SECTION 7 — SUPPLY SIDE & AGRICULTURAL CONSTRAINTS

- **Capacity (Reported):** Installed ethanol capacity ~**1,800–2,000 crore litres (18–20 bn L)** by 2025, from <2 bn L in 2014. For ESY 2025-26, OMCs received offers of Cartoq **1,776 cr L** against a ~1,050 cr L requirement — **structural oversupply.**
- **Feedstock mix shift (Reported — the key structural change):** Maize is now the **No.1 feedstock** at ~48–51% of supply (ESY 2024-25); grain-based ~69% of total, sugarcane ~31%. This reverses the original molasses-dominated model. Maize ethanol output rose ~9x (315 mn L in ESY22-23 → 2.86 bn L in ESY23-24).
- **Feedstock prices (Reported, ESY 2024-25):** Sugarcane juice/syrup ₹65.61/L; Substack B-heavy molasses ₹60.73/L; C-heavy molasses ₹57.97/L; damaged foodgrains ₹64/L; **maize ₹71.86/L (highest)**; surplus FCI rice ₹60.32/L (ESY 2025-26).
- **Water intensity (Reported — NITI Aayog Roadmap, June 2021):** "One litre of ethanol from sugar requires about **2,860 litres of water**" (sourced to University of Twente, 2009); the same report states "sugarcane and paddy combined use 70% of irrigation water of the country." Maize ~4,670 L/L and rice ~10,790 L/L (Food Secretary, India Today). CSTEP estimates annual irrigation demand could rise ~50 billion m³ by 2070.
- **Land (Reported/Estimated):** NITI estimated ~4.8 million ha of additional maize needed for E20. Land used for ethanol feedstock rose ~8x in four years (Arcus/Reuters).
- **Food-vs-fuel (Reported):** Maize for ethanol surged 0.8 MT (ESY22-23) → 12.7 MT (ESY24-25); India flipped from maize exporter (3.7 MT, 2021-22) to importer; maize prices rose from ₹14–15k/t to ₹24–25k/t; the poultry/feed sector (60–70% of maize use) was squeezed; ~5.2 MT FCI rice diverted in 2024-25. USDA The DDGS byproduct partly offsets feed loss but undercuts soymeal, hurting soybean farmers.
- **2G cellulosic (Reported):** IOCL Panipat (₹900+ cr, 100 KLPD / ~3 cr L/yr from rice

straw) Prime Minister of India is **operating below design capacity.** Only ~2 of 12 planned 2G plants are running; India's pilot 2G capacity ~32 ML/yr. USDA PM JI-VAN scheme allocated ₹1,969.5 cr for 12 projects. Ministry of Petroleum and Natural G

- **Scalability (Opinion):** Supply is no longer the binding constraint — demand is. But the *quality* of supply (maize/rice-based) erodes the program's environmental and food-security rationale, and 2G remains a chronic underperformer despite a decade of subsidies.

**Chart idea 7A:** Feedstock mix evolution (stacked area), ESY 2018→2026: C-molasses / B-molasses / cane juice / maize / FCI rice / damaged grain. **Chart idea 7B:** Water footprint per litre of ethanol by feedstock (bar): rice 10,790 / maize 4,670 / sugarcane 2,860. **Chart idea 7C:** Installed capacity vs OMC requirement vs offered volume (bar, ESY 2025-26: ~1,800 cap / ~1,050 req / 1,776 offered).

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## SECTION 8 — ECONOMICS & PRICING

- **Administered pricing (Reported):** Government sets ex-mill prices for molasses/cane-juice ethanol; OMCs set grain-ethanol prices. National Single Window System Prices are **delinked from crude** and revised by feedstock annually. Ethanol procurement (₹57–72/L) frequently **exceeds petrol's pre-tax base price** (~₹52.83/L in Delhi per critics) Substack — i.e., ethanol is NOT cheaper than petrol before taxes.
- **Crude break-even (Reported/Estimated):** Classic literature places corn ethanol's competitiveness (ex-subsidy) with gasoline above ~$50–60/bbl crude. uspto India's administered system means ethanol economics are politically set, not market-cleared. Industry estimate: every $1/bbl crude rise adds ~$2 bn to India's import bill Khaitan Bioenergy — the macro-hedge rationale.
- **Can ethanol compete WITHOUT subsidies? (Opinion): No, not currently in India.** Evidence: import restrictions (no foreign competition), guaranteed OMC offtake, interest subvention, Substack concessional 5% GST. It is a textbook protected/administered market. Economics improve at high crude ($90+) and worsen at low crude ($50–60).
- **Sensitivity framework (Estimated):**
    - **Low crude (~$50–60):** Ethanol uneconomic vs petrol; mandate-dependent; sugar/grain producers squeezed; political pressure not to raise ethanol prices.
    - **Base crude (~$70–80):** Roughly competitive on a policy-supported basis; current equilibrium.
    - **High crude (~$90–110):** Macro-attractive (forex savings); strong tailwind for the

theme.

- **Vs hybrids/EVs (Opinion):** On lifecycle cost per km, EVs (2W/3W) already win where charging exists; ethanol-ICE does not beat EVs on running cost given the energy-density penalty and same-or-higher pump price. Ethanol's edge is **capital efficiency** (reuses fleet/infrastructure) and **farm politics**, not consumer economics.

**Chart idea 8A:** Ethanol price by feedstock vs petrol pre-tax base (bar), ESY 2024-25.
**Chart idea 8B:** Crude-price sensitivity matrix (low/base/high) mapped to ethanol attractiveness and import-bill savings.

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## SECTION 9 — ETHANOL VS OTHER POWERTRAINS

| Dimension | Ethanol ICE (E20) | Flex-fuel (E85/E100) | Hybrid (HEV) | Battery EV |
|---|---|---|---|---|
| Upfront cost | Lowest (≈petrol) | Modest premium; **18–40% GST** | Premium | High; **5% GST** |
| Running cost/km | Slightly worse than petrol (energy penalty, same price) | Depends on ethanol price gap | Better | Best (where charging) |
| WTW efficiency | Low (ICE) | Low (ICE) | Medium | ~19.6% WTW on 2023 India coal-heavy grid (improves as grid greens) |
| Lifecycle $CO_2$ | Feedstock-dependent | Feedstock-dependent | Lower than ICE | India 2021: ~19–34% lower than petrol; improves with >50% non-fossil installed capacity |
| Infrastructure | Reuses existing | Needs E85/E100 pumps (sparse) | Reuses existing | Needs charging buildout |
| | Capped by | Capped by | | |

| | | | | |
|---|---|---|---|---|
| Scalability | feedstock | feedstock + GST | High | High |

- **Segment verdict (Opinion):**
  - **Two-wheelers (61% of petrol):** EVs likely win medium-term (TCO, simple charging/swap); ethanol-ICE bridges near-term.
  - **Passenger vehicles:** Hybrids + E20-ICE near-term; EVs gain post-2030; flex-fuel a niche unless GST is corrected.
  - **Commercial/long-haul:** Diesel-dominated. Since ethanol is a *petrol* substitute, it is **largely irrelevant to the ~91 MMT diesel pool** — a crucial structural limiter on ethanol's total oil-import impact.

**Chart idea 9A:** Powertrain 5-year TCO bars per segment (2W/PV) for ICE-E20 / HEV / BEV.
**Chart idea 9B:** WTW $CO_2$ (g/km) by powertrain on current India grid vs a 2030 greener grid.

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## SECTION 10 — INVESTMENT LENS (thematic, category-level; no tickers)

**Category winners:**

- **Sugar companies / integrated mills:** Structural re-rating — ethanol gives predictable, government-backed cash flows, faster OMC payments, Goodwill's Blog margin expansion (cane-juice/B-heavy ethanol > sugar margins), and balance-sheet deleveraging. Goodwill's Blog Best positioned: feedstock-flexible, vertically integrated, high distillery capacity. Goodwill's Blog KEY RISK: cane-juice diversion caps, monsoon/drought.
- **Grain processors / maize-based distillers:** Fastest-growing feedstock; decoupled from the sugar cycle; beneficiary of a >₹40,000 cr investment wave. Substack RISK: maize price inflation, food-feed backlash, GM-import policy uncertainty.
- **Engineering/EPC & distillery/2G technology licensors:** Capex beneficiaries of the capacity build-out.

**Mixed/neutral:**

- **OMCs:** Forced buyers at administered prices (largely margin-neutral pass-through); blending is policy compliance, not a profit center. Ethanol costs above petrol's base can pressure marketing margins when not fully passed through.

- **Auto OEMs:** E20 compliance largely absorbed; modest cost. Flex-fuel is a strategic option but GST-constrained. OEMs are hedged across ICE/hybrid/EV.

**Potential losers / at-risk:**

- **Pure-play sugarcane-only mills** in water-stressed belts (drought exposure).
- **Poultry/animal-feed sector** (maize cost inflation) — a *downstream loser* of the ethanol pivot.
- **Legacy-vehicle owners / aftermarket** (compatibility costs).
- **EV ecosystem (short-term):** The ethanol push is a partial policy hedge that could blunt the urgency of EV incentives — but EVs remain the long-run structural winner in 2W/3W.

**Key variables to monitor (10–20 yr):**

1. Blend-mandate trajectory beyond E20 (E25/E27/E30 notifications).
2. Flex-fuel GST rationalization (18–40% → toward 5–12%?).
3. Ethanol oversupply vs demand gap (capacity utilization).
4. Feedstock mix (maize/rice share = food-security risk gauge).
5. Crude oil price (the macro tailwind/headwind).
6. EV adoption curve in 2W/3W.
7. Water-stress regulation / NITI interventions on cane.
8. 2G ethanol commercial-viability breakthrough.

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## SECTION 11 — METHODOLOGY, SOURCES & CONTRADICTIONS

**Authoritative sources used / cross-referenced:** NITI Aayog "Roadmap for Ethanol Blending in India 2020-25"; PPAC consumption/import data; MoPNG/PIB releases; SIAM/PPAC-Nielsen sectoral study; ARAI/IOC R&D/IIP (via MoPNG statements); USDA FAS Biofuels & Sugar Annuals (India, Brazil, Thailand); IEA & IEA-AMF fuel-property data; US EPA (Brazil LCA); US DOE/AFDC; ePURE / European Parliament / RED III documents; peer-reviewed LCAs (ScienceDirect WTW-India studies); ISMA/AIDA; CSEP; The India Forum; S&P Global Platts.

**Note on Brazil LCA (Reported):** US EPA (2010 RFS2 finalization) found Brazilian sugarcane ethanol reduces GHG emissions vs gasoline by **61%**, using a 30-year payback for indirect land-use-change (ILUC) emissions; EPA estimated ILUC at ~4.5 g $CO_2e$/MJ (per a 2023

GREET/RenovaBio analysis in *Environmental Science & Technology*).

**Identified contradictions in the ethanol narrative (critical for an institutional view):**

1. **Farmer-income vs food-security:** The maize/rice pivot boosts cane/maize farmers but inflates feed costs, hurts poultry & soybean farmers, and triggers imports.
2. **Energy-security vs water-stress:** Domestic fuel reduces oil imports but consumes 2,860–10,790 L of water per L of ethanol in a water-stressed nation where cane+paddy already use ~70% of irrigation water.
3. **Decarbonization vs land-use/feedstock emissions:** Sugarcane ethanol cuts $CO_2$, but maize ethanol is ~2x its GHG and rice-based is far worse; lifecycle claims are entirely feedstock-dependent.
4. **Original purpose vs current reality:** The program began to clear a *sugar surplus*; it is now a *grain-based* industry — undermining the waste-to-value logic.
5. **"Marginal" mileage drop (official 1–6%) vs user-reported 5–12%, with ~50% (Feb 2026 survey) to ~80% (Oct 2025 survey) of older-vehicle owners reporting decline.**
6. **NITI recommended a lower retail price for blended fuel; policy charges the same/higher** — the consumer effectively subsidizes the producer.
7. **Oversupply vs the higher-blend push:** Capacity already exceeds E20 demand, yet the lobby pushes E25–E30 requiring substantial fresh capex and subsidies Substack — which critics characterize as crony capitalism.

**Data flags (transparency):** Government benefit claims (forex saved, $CO_2$ avoided, farmer payments) are *reported/self-stated* and rise across successive statements (₹1.44 lakh cr / 736 lakh MT $CO_2$ in Aug 2025 → ₹1.84 lakh cr / 909 lakh MT $CO_2$ by June 2026) — not independently audited. The E20-nationwide effective date straddles **2025** (20% blending milestone, reached October 2025) and **April 2026** (E20 as the standard petrol grade at min RON 95, with E0 largely phased out). Crude import bill: **$137 bn gross (PPAC)** vs ICRA's **~$101–104 bn net** estimate. Several flex-fuel passenger-car "launches" (e.g., Maruti Wagon R flex, mid-2026) are reported by auto-media and should be corroborated against official OEM/PIB announcements before being treated as definitive; flex-fuel two-wheelers (Hero HF Deluxe/Splendor+, Suzuki Gixxer SF 250, TVS Apache RTR 200 E100) are in market, while flex-fuel cars (Toyota Innova HyCross, Tata Punch, Hyundai Creta) remain largely at prototype/early-production stage.

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## BOTTOM-LINE SUMMARY (Opinion, for the report author)

Ethanol in India is a **real, durable, policy-driven theme through ~E20–E27 over 10–15 years** — genuinely accretive to integrated sugar mills and grain distillers, neutral for OMCs, and modestly positive/strategic for OEMs. But it is **not** the long-term winner of India's transport-decarbonization race: it cannot touch the diesel pool, faces a hard ~20–30% volumetric blend ceiling, depends on administered prices and subsidies, and carries unresolved water/food contradictions. The theme is best traded as a **near-to-medium-term agri-energy capacity-and-margins story with a clear expiry risk as EVs scale**, monitored against the eight variables in Section 10. Treat it as a *policy-supported structural moderator of petrol demand*, not an enduring substitute for it.